



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

APACHE BEAM PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is Apache Beam and what problem in Data Engineering does it solve?

Threat Model

Q6 How does Apache Beam handle reliability and high availability?

Observability

Q2 What are the core components or concepts in Apache Beam?

Architecture

Q7 What observability does Apache Beam provide out of the box?

Lifecycle

Q3 How does Apache Beam compare to its main alternatives?

Configuration

Q8 What are common performance bottlenecks in Apache Beam?

Compliance

Q4 How do you get started with Apache Beam for a new project?

Hardening

Q9 What security best practices apply when running Apache Beam?

Testing

Q5 What configuration options most affect Apache Beam's behavior in production?

SSO / IdP

Q10 What mistakes do teams commonly make when adopting Apache Beam?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Apache Beam is a tool or

Mitigation

Apache Beam is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A6 Apache Beam provides HA through leader

Observability

Apache Beam provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Apache Beam has key building blocks

Primitives

Apache Beam has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Apache Beam exposes metrics (Prometheus)

Automation

Apache Beam exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Apache Beam trades off simplicity against

Hardening

Apache Beam trades off simplicity against feature richness differently than its alternatives. Choose Apache Beam when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Apache Beam via its package

Gotchas

Install Apache Beam via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Apache Beam with the principle

Assessment

Run Apache Beam with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.